

## §2.6

# Order of Operations

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One agreed order, so that every expression has exactly one value

# Today's Goal

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After completing this section, you should be able to:

- Explain why a single agreed-upon convention is needed so that every numerical expression has one unambiguous value
- Apply the order of operations—grouping symbols, exponents, multiplication and division, addition and subtraction—to evaluate numerical expressions, including expressions with nested grouping symbols
- Avoid the common misconception that multiplication always precedes division (or addition always precedes subtraction) by evaluating same-tier operations from left to right

# Learning Target 8

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## LT.8 Order of Operations

I can apply the order of operations correctly to evaluate a numerical expression.

# Four Students, Four Answers

Four students evaluate  $12 + 24 \div 6 \times 2$ . What did each one do first? Which is right?

**Student 1.**

$$\begin{aligned} & 36 \div 6 \times 2 \\ & = 6 \times 2 = 12 \end{aligned}$$

**Student 2.**

$$\begin{aligned} & 36 \div 6 \times 2 \\ & = 36 \div 12 = 3 \end{aligned}$$

**Student 3.**

$$\begin{aligned} & 12 + 24 \div 12 \\ & = 12 + 2 = 14 \end{aligned}$$

**Student 4.**

$$\begin{aligned} & 12 + 4 \times 2 \\ & = 12 + 8 = 20 \end{aligned}$$

# One Agreed Order

## Definition 2.6.1

The *order of operations* is the agreed-upon convention for the sequence in which the operations in a numerical expression are evaluated, so that every expression has a single, unambiguous value.

1. *Grouping symbols.* Evaluate anything inside grouping symbols (parentheses, brackets) first. When grouping symbols are nested within one another, start with the innermost and work outward.
2. *Exponents.* Evaluate powers.

# One Agreed Order, Continued

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## Definition 2.6.1, continued

3. *Multiplication and division.* Perform multiplications and divisions as they appear from left to right.
4. *Addition and subtraction.* Perform additions and subtractions as they appear from left to right.

# PEMDAS and GEMDAS

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## PEMDAS

**P**arentheses

**E**xponents

**M**ultiplication and **D**ivision (left to right)

**A**ddition and **S**ubtraction (left to right)

## GEMDAS

**G**rouping Symbols

**E**xponents

**M**ultiplication and **D**ivision (left to right)

**A**ddition and **S**ubtraction (left to right)

GEMDAS reminds us that *any* grouping symbol comes first, not just parentheses.

# The Left-to-Right Trap

## Watch out

Multiplication does not always come before division, and addition does not always come before subtraction.

$$\begin{aligned} & 12 + 24 \div 6 \times 2 \\ = & 12 + 4 \times 2 && \text{division first, it is leftmost} \\ = & 12 + 8 && \text{then multiplication} \\ = & 20 \end{aligned}$$

Student 4 is right.

# Working From the Inside Out

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Evaluate  $3 \times (8 - [10 - (4 + 3)])$ . Start from the inside and work out.

$$\begin{aligned} & 3 \times (8 - [10 - (4 + 3)]) \\ = & 3 \times (8 - [10 - 7]) && \text{innermost group: } (4 + 3) \\ = & 3 \times (8 - 3) && \text{next group: } [10 - 7] \\ = & 3 \times 5 && \text{last group: } (8 - 3) \\ = & 15 \end{aligned}$$

# No Grouping Symbols At All

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Evaluate  $24 \div 6 \times 3 + 5$ .

$$\begin{aligned} & 24 \div 6 \times 3 + 5 \\ &= 4 \times 3 + 5 && \text{division first, it is leftmost} \\ &= 12 + 5 && \text{then multiplication} \\ &= 17 \end{aligned}$$

## Two Groups, Side by Side

Evaluate  $(2 + 6) \times [3^2 - 15 \div 5]^2 + 4$ .

$$\begin{aligned} & (2 + 6) \times [3^2 - 15 \div 5]^2 + 4 \\ = & 8 \times [3^2 - 15 \div 5]^2 + 4 \\ = & 8 \times [9 - 15 \div 5]^2 + 4 \\ = & 8 \times [9 - 3]^2 + 4 \\ = & 8 \times 6^2 + 4 \\ = & 8 \times 36 + 4 \\ = & 288 + 4 \\ = & 292 \end{aligned}$$

leftmost group  
exponent inside  
division inside  
the group  
exponent  
multiplication

# Make the Target Number

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**Group work.** You are given 2, 3, 5, and 8.

**Use each number once, with  $+$ ,  $-$ ,  $\times$ ,  $\div$  and grouping symbols, to make 48.**

Stuck? These numbers are small, so adding them all is not enough. Which two numbers multiply to 48?

# Two Ways to Hit 48

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Pairs that multiply to 48:

1.  $2 \times 24$

2.  $3 \times 16$

3.  $4 \times 12$

4.  $6 \times 8$

Pair (4) works: make 6 from 5, 3, and 2.

$$\begin{aligned} & 8 \times (5 + 3 - 2) \\ &= 8 \times 6 = 48 \end{aligned}$$

$$\begin{aligned} & (8 - 2) \times (5 + 3) \\ &= 6 \times 8 = 48 \end{aligned}$$

Every expression you test makes you apply the order of operations again.

# One Card, Many Puzzles



Build the target 4 from the five edge numbers. Each number may be used only once.

- **Easy.** Use two or more.
- **Medium.** Use three or more.
- **Hard.** Use four or more.
- **Monstrous.** Use all five.

Can you do it with exactly two numbers? Three? Four? Five?

# One Question to Leave With

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**Exponents sit above  
multiplication in the  
order of operations.  
What does that buy us?**

An exponent is shorthand for a whole run of multiplications. Doing it first keeps

$2 \times 3^2$  meaning  $2 \times 9$ , not  $(2 \times 3)^2$ .